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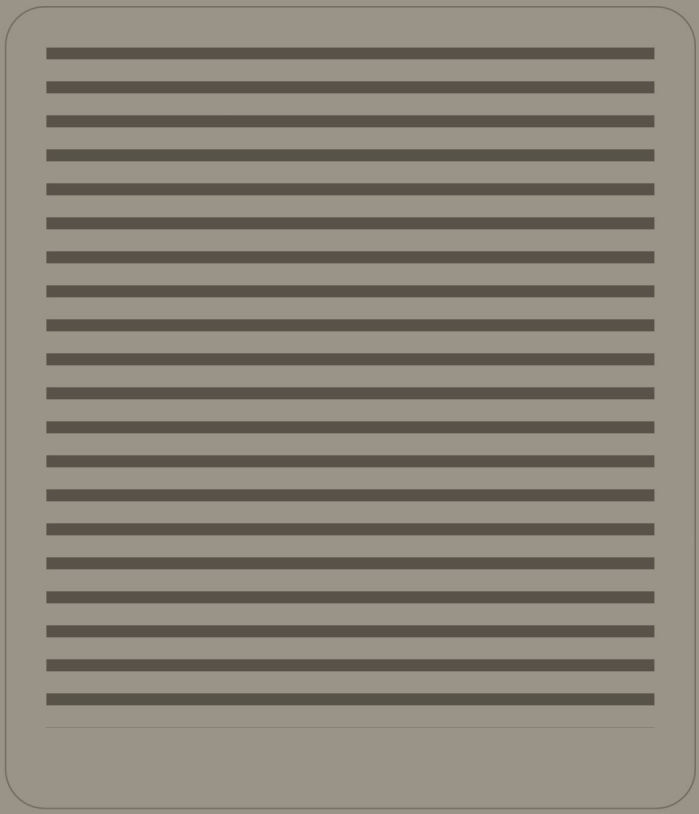
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A process is a program in execution. Each process has its own address space, registers, and open files, which the kernel tracks in a

Ask AI | Generate Diagram

The scheduler saves the process's state, selects the next ready process, and restores that process's saved state before resuming it. This sequence is called a context switch.

Switching is not free: the CPU does no useful work while state is saved and restored, so the cost is pure overhead.

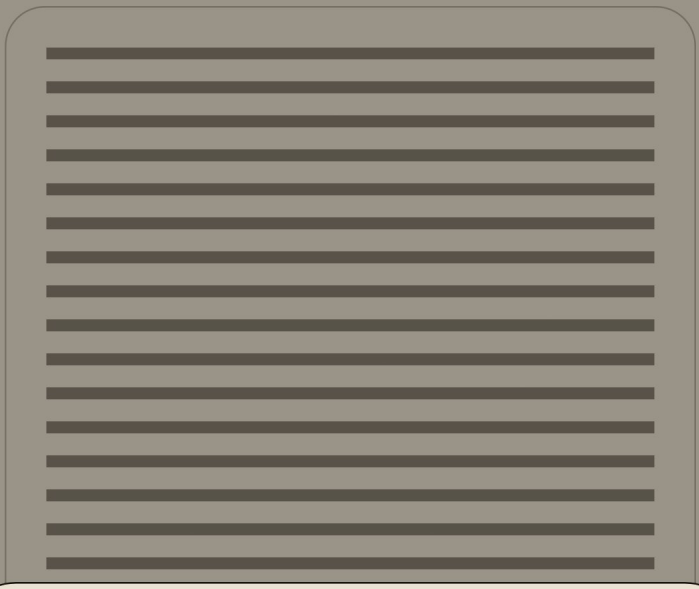


ASK ABOUT THIS PASSAGE

“The scheduler saves the process’s state, selects the next ready process, and restores that process’s saved state before resuming it.”

Explain | Hint first

Why does it restore the state?



ANSWER

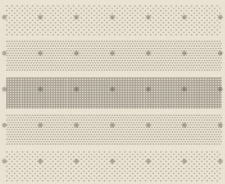
“The scheduler saves the process’s state, selects the next ready process, and restores that process’s saved state before resuming it.”

That sequence is a **context switch**. The kernel copies the running process’s registers and program counter into its process control block, picks another ready process, then loads that process’s saved values back into the CPU.

Restoring state is what lets a process resume exactly where it stopped, as if it had never been interrupted.

Generate Diagram

Answer | Diagram

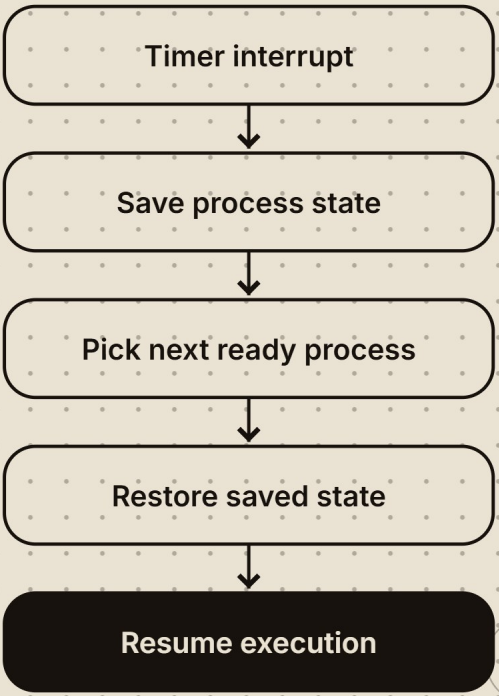


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Answer | Diagram

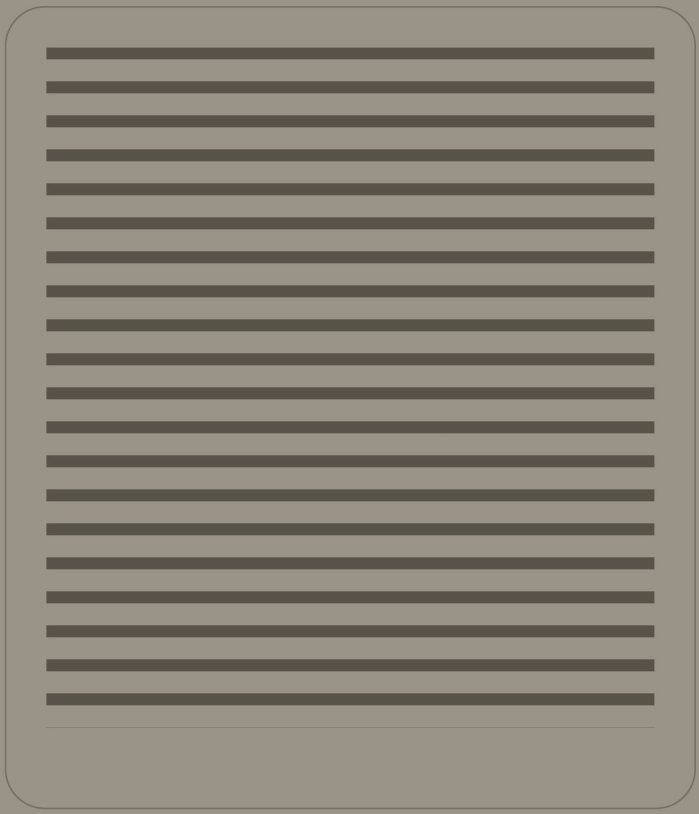


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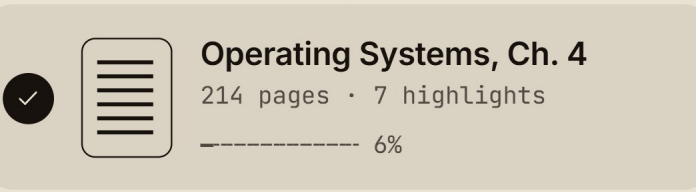
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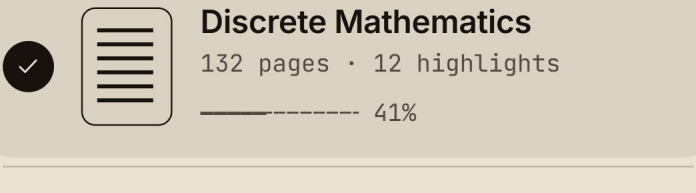
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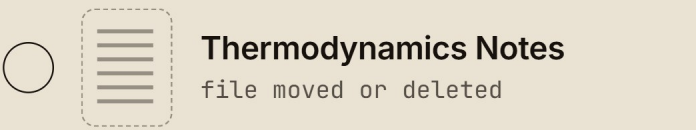
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A process is a program in execution.

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Each process has its own address space, registers, and open files, which the kernel tracks in a process control block.

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